

JYOT SHAH

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PROFILE

Computer Science undergraduate specializing in Machine Learning and full-stack software development. Strong foundation in Data Structures and Algorithms, Object Oriented Programming, and Database Systems, with experience in building and deploying ML models and web-based applications. Proficient in end-to-end system development, model training, API integration, and scalable backend design, with a focus on solving real-world engineering problems.

EDUCATION

Manipal University Jaipur

2023 – 2027

Bachelor of Technology - *Computer Science & Engineering (Specialization in AI & ML)*

Current **CGPA: 9.25**

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, C++

Frontend: React.js, HTML, CSS, EJS

Backend & Databases: Node.js, Express.js, Flask, RESTful APIs, MongoDB, MySQL

ML Frameworks: TensorFlow, Scikit-learn, Keras, OpenCV, Pillow, NumPy, Pandas

Concepts: DSA, OOP, Operating Systems, DBMS, Computer Networks, MVC Architecture, NLP, Computer Vision

Soft Skills: Problem-Solving, Team Collaboration, Technical Communication

PROJECTS

Plant Leaf Disease Detection & Treatment System ([Code](#)) | *YOLOv11, Flask, Gemini*

2025

- Built an end-to-end plant disease detection system using a custom-trained YOLOv11 model on an augmented PlantDoc dataset, achieving ~96% validation accuracy with optimized preprocessing and real-time inference deployed via Flask.
- Integrated a Gemini 2.5 Flash powered treatment chatbot and a responsive frontend, enabling instant annotated predictions and context-aware treatment recommendations, improving diagnostic interaction speed by ~40%.

InfraGhost AI - Infrastructure Reality Verification System ([Code](#)) ([Live](#)) | *MERN Stack, Gemini Vision AI*

2026

- Built a secure full-stack AI platform using Gemini Vision to analyze geotagged infrastructure reports and compute a 0–100 Ghost Score for automated asset classification; client-side image compression reduced payload size by ~60%, improving AI response latency by ~35%.
- Developed a hardened Express.js backend with rate limiting and input validation, integrated MongoDB storage, and designed a real-time Mapbox analytics dashboard with statistics and PDF reporting for scalable authority-level monitoring.

WanderLust – Airbnb-Style Web App ([Code](#)) ([Live](#)) | *MERN Stack, Render*

2025

- Developed a full-stack Airbnb clone using Node.js, Express.js, MongoDB, and EJS, implementing secure authentication (Passport.js), session management, and complete CRUD operations for listings and reviews, improving backend request handling efficiency by ~30% through optimized routing and middlewares.
- Integrated Cloudinary for scalable image uploads and Mapbox for dynamic location visualization, while enforcing robust server-side validation with Joi, resulting in a 40% reduction in invalid form submissions and a smoother user experience.

RESEARCH

Hierarchical Vision Transformer Ensemble for Robust Plant Leaf Disease Classification

Ongoing

- Curating and training on diverse plant leaf datasets (80K+ images, 30+ classes), using class-balanced sampling and controlled augmentations to mitigate domain shift; achieving ~97% validation accuracy with improved dataset stability.
- Building a hierarchical ViT with parent-child attention, combined with lightweight ensembling and a hybrid loss (CE + Focal + adaptive margin), yielding 4–6% F1 gains over baseline CNN/ViT models while remaining efficient on limited GPU setups.

CERTIFICATIONS

• IBM AI Engineering Degree Certificate (*Coursera*)

• Sigma 8.0 – MERN Stack + DSA (*Apna College*)

ACHIEVEMENTS

Honoured with the Dean's List award for academic excellence in the 1st, 2nd, and 4th semesters, consistently ranking among the top-performing students in the Department of AIML at Manipal University Jaipur.